

## Layout System for Individuals

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**L**ayout System for Individuals, popularly known as LASI, is a PC based computer aided design (CAD) software primarily created for Windows operating system. As the name defines, LASI is a layout and design tool for full-custom very large scale integration (VLSI) micro-electronic circuit designs that include MEMS, ICs, discrete devices, personal computer (PC) boards, schematics, different project documentation designs and other technologies.

LASI can create and arrange polygons corresponding to regions like

## Quick view

- Version: 7.0.86
- Licence: Open source
- Operating system: Windows
- Key features:
  - Circuit construction is done through a hierarchy of cells
  - Has an inbuilt utility for DRC of the circuit layout so designed

metal interconnect and drain to create each device in the IC or any other design, which makes it a versatile and a general-purpose tool for design and layout. It is a free software tool and is used mainly for educational purposes.

## LASI and operating systems

Although the current version of this tool can work on all different versions of Windows (XP, Vista, or Windows 7 or 8), its initial version was written in MS-DOS. This initial version was the simplest version available till the advent of Windows operating system (OS). Now, this does not mean that you cannot work with LASI on any other OS. Remember, technology has improved and is improving every minute. You can even use a virtual Windows OS on your main OS to install and use LASI.

## Competitive tools

- CLEwin      ➤ Layout

Another benefit that LASI offers is portability. You can install this software either on a flash drive or any removable drive and work on drawings saved on the same drive. Portability of tools allows you to work on any PC. Layouts thus created can be transferred to other systems in the most-commonly-used graphic format in electronics called graphical database system (GDS).

## Cell transportation

LASI follows hierarchy in layout and design. It builds drawings from hierarchical cells that are stored as individual transportable layout cell (TLC) files or transportable LASI drawing (TLD) files. These drawings or layouts are saved as text files with TLC extensions. These TLC files contain references of other cells but not actual layout of the referenced cell.

In order to find the hierarchy of cells, TLD files are used. You can generate a TLD file using Export command by selecting the name of the

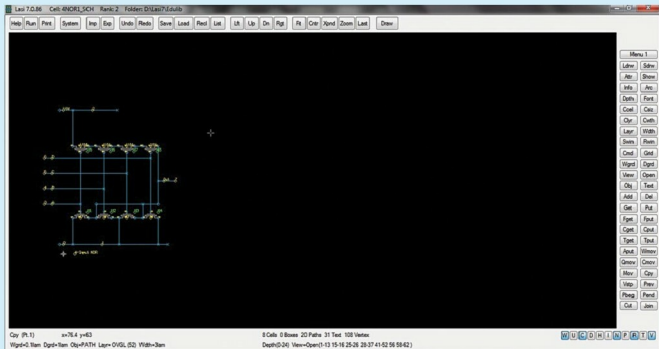


Fig. 1: NOR gate representation